EXPERIMENTAL ANALYSIS OF CARBON DIOXIDE UPTAKE AND OXYGEN RELEASE IN PLANTS

PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the course

of

GEA1401 – ENVIRONMENTAL SCIENCE (TCPR)

IN

DEPARTMENT OF

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM,TRICHY

ANNAUNIVERSITY

CHENNAI 600 025

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Under the Guidance of

Mr. Dr .K. LENIN , M.E., Ph.D.,

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BONAFIDE CERTIFICATE

Certified that this project report titled **"EXPERIMENTAL ANALYSIS OF CARBON DIOXIDE UPTAKE AND OXYGEN RELEASE IN PLANTS"** is the Bonafide work of **PRADHAP R (8115U23ME035), SARATHIKANNAN R M (8115U23ME041), STEVE JASON A (8115U23ME049), VISHWA C (8115U23ME061),** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Under
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We jointly declare that the project report on **"EXPERIMENTAL ANALYSIS OF CARBON DIOXIDE UPTAKE AND OXYGEN RELEASEIN PLANTS"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to "ANNA UNIVERSITY CHENNAI" for the requirement of Degree of BACHELOR OF ENGINEERING. This project report is submitted on the partial fulfillment of the requirement of the award of degree of BACHELOR OF ENGINEERING.

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DEPARTMENT OF MECHANICAL ENGINEERING
Program: Bachelor of Engineering

VISION & MISSION OF THE COLLEGE

VISION:

- To achieve a prominent position among the top technical institutions

MISSION:

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To provide education that can be used as a tool to transform the students.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

MISSION:

- To impart quality professional education through state-of-the-art-infrastructure and resources.
- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

DEPARTMENT OF MECHANICAL ENGINEERING

Program: Bachelor of Engineering

Program Educational Objectives :(PEOs)

Graduates of Mechanical Engineering shall:

1. Have Successful careers in industries or opt for higher studies and research.

2. Analyze, synthesize and design mechanical and allied products addressing societal issues.

3. Exhibit technical expertise, communication skills, team work with professionalism with ethical values.

Program Specific Outcomes (PSOs)

Students shall:

PSO1: Identify, formulate and solve engineering problems in three core streams of Mechanical Engineering, i.e., Design, Thermal and Manufacturing Engineering

PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

PROGRAM OUTCOMES

PROGRAM OUTCOMES (PO's)	
PO 1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledg and research methods including design of experiments, analysis an interpretation of data, and synthesis of the information to provide vali conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Program Outcomes (POs) Applied in This Project		
PO No.	**Program Outcome**	**Applied in Project**
PO 1	**Engineering Knowledge** Applies engineering, electronics, and scientific knowledge to study carbon dioxide uptake and oxygen release in plants.	✓
PO 2	**Problem Analysis** Identifies and analyzes changes in CO_2 and O_2 concentrations during photosynthesis using sensor readings.	✓
PO 3	**Design/Development of Solutions** Develops a monitoring system using NodeMCU ESP8266, gas sensors, and an LCD display.	✓
PO 4	**Conduct Investigations of Complex Problems** Conducts experiments under different lighting conditions and analyzes the collected gas concentration data.	✓
PO 5	**Modern Tool Usage** Uses the NodeMCU ESP8266, CO_2 sensor, oxygen sensor, and LCD to monitor and display experimental results.	✓
PO 6	**The Engineer and Society** Supports awareness of plant functions and their importance in maintaining environmental balance.	✓
PO 7	**Environment and Sustainability** Demonstrates the role of plants in absorbing CO_2 and releasing oxygen, contributing to environmental sustainability.	✓
PO 8	**Ethics** Encourages accurate data recording, responsible experimentation, and honest reporting of results.	✓
PO 9	**Individual and Team Work** Encourages teamwork in circuit assembly, sensor calibration, experimentation, and result analysis.	✓
PO 10	**Communication** Improves the ability to document experimental procedures, present sensor readings, and communicate project findings.	✓
PO 11	**Project Management and Finance** Develops skills in component selection, budget planning, resource management, and timely completion of the project.	✓

SUSTAINABLE DEVELOPMENT GOALS



PRIMARY SDG'S USED IN THE PROJECT



SDG 13: Climate Action:
The project studies carbon dioxide uptake by plants and oxygen release during photosynthesis, supporting awareness of the carbon cycle and the role of vegetation in climate-related environmental processes.

SECONDARY SDG'S USED IN THE PROJECT



SDG 15: Life on Land:
Monitoring plant activity helps improve understanding of plant health and the importance of protecting terrestrial ecosystems.



SDG 9: Industry, Innovation and Infrastructure:
The project combines a NodeMCU ESP8266, CO_2 sensor, oxygen sensor and LCD display to create an affordable experimental monitoring system.



SDG 12: Responsible Consumption and Production:
Low-cost sensing and reusable electronic components support efficient experimental resource use and can encourage environmentally responsible monitoring.

ABSTRACT

The project titled “Experimental Analysis of Carbon Dioxide Uptake and Oxygen Release in Plants” focuses on studying the photosynthesis process and understanding the role of plants in maintaining atmospheric balance. Plants absorb carbon dioxide from the surrounding environment through tiny openings called stomata and use sunlight, water, and chlorophyll to produce glucose while releasing oxygen as a by-product. This project aims to experimentally measure changes in carbon dioxide and oxygen concentrations during photosynthesis using suitable sensors and an experimental chamber.

A carbon dioxide sensor is used to monitor the reduction in carbon dioxide concentration, while an oxygen sensor measures changes in oxygen levels during the experiment. An Arduino or ESP32 microcontroller can be used to collect sensor readings and process the measured data for analysis. A light source provides the required illumination, and a temperature sensor monitors environmental conditions that may influence photosynthetic activity. The experiment can be conducted under different light intensities and environmental conditions to investigate their effects on carbon dioxide uptake and oxygen release.

The collected data is recorded, compared, and analyzed to understand the relationship between photosynthesis and environmental factors. Proper experimental controls are necessary to distinguish changes caused by plant activity from variations due to leakage, respiration, and sensor limitations. This study helps improve understanding of plant physiology, supports the development of sensor-based environmental monitoring systems, and highlights the importance of vegetation in absorbing atmospheric carbon dioxide and releasing oxygen.

Keywords:

photosynthesis, carbon dioxide, oxygen, NodeMCU ESP8266, NDIR sensor, gas exchange.

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LIST OF ABBREVATION

S.NO	ABBREVATION	WORD
1	OXYGEN SENSOR	O2
2	CARBON DI OXIDE	CO2
3	MICROCONTROLLER	MCU

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Life on Earth depends on the continuous exchange of gases between living organisms and the atmosphere. Animals and microorganisms consume oxygen (O_2) and release carbon dioxide (CO_2) through respiration, whereas green plants do the opposite during daylight: they absorb CO_2 and release O_2 through photosynthesis. This balance has kept the composition of air relatively stable for millions of years, with oxygen near 20.9 % and carbon dioxide at a trace level of a few hundred parts per million (ppm).

Since the industrial revolution, burning of fossil fuels, deforestation and rapid urbanisation have pushed atmospheric CO_2 well above its pre-industrial level of about 280 ppm. In this context the role of plants as natural carbon sinks and oxygen sources has become a central topic in Environmental Studies (EVS). Understanding how much CO_2 a plant takes up, and how much O_2 it returns to the air, requires measurement and not only textbook statements. This project therefore develops a small, low-cost experimental set-up that measures both gases around a living plant.

1.2 Photosynthesis and Gas Exchange in Plants

Photosynthesis is the process in which chlorophyll-containing cells convert light energy into chemical energy. The overall reaction is $6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$. Carbon dioxide enters the leaf mainly through microscopic pores called stomata and is fixed in the chloroplasts by the Calvin cycle, while oxygen produced by the splitting of water in the light reactions diffuses out through the same pores.

At the same time, plants respire continuously. In darkness, photosynthesis stops but respiration goes on, so the plant becomes a net source of CO_2 and a net consumer of O_2. The net gas exchange that an observer measures is therefore the difference between photosynthesis and respiration, and it changes with light intensity, temperature, humidity, CO_2 concentration and the health of the plant.

1.3 Importance of Monitoring CO_2 and O_2

Measuring gas concentrations around a plant converts an invisible process into numbers that can be plotted and compared. Such data help students and researchers to estimate the photosynthetic rate, to compare species, and to understand how factors such as light and enclosure volume affect air quality. Indoor air quality is also a growing concern: closed rooms can accumulate CO_2 above 1000 ppm, causing drowsiness and reduced concentration, and indoor plants are often promoted as a remedy.

Monitoring both gases together is more informative than monitoring either alone. According to the photosynthetic equation, one molecule of O_2 is released for every molecule of CO_2 fixed, so the two readings should move in opposite directions and in nearly equal magnitude. Any strong deviation points to sensor drift, leakage of the chamber or an unusual physiological state of the plant.

1.4 Role of IoT and Sensors in Environmental Monitoring

Low-cost microcontrollers and gas sensors have made environmental monitoring accessible to educational laboratories. The NodeMCU ESP8266 board combines a 32-bit processor with built-in Wi-Fi, allowing readings to be displayed locally and also streamed to a computer or cloud dashboard. Non-dispersive infrared (NDIR) CO_2 sensors give stable and selective measurements, and electrochemical oxygen sensors provide a linear voltage output proportional to the partial pressure of oxygen.

Using such components, a compact data-logging system can be built for a few thousand rupees, in contrast to commercial leaf-chamber photosynthesis systems that cost several lakhs.

1.5 Scope of the Project

The scope of this project is the design, fabrication and testing of a transparent, sealed plant chamber fitted with a CO_2 sensor, an O_2 sensor, a NodeMCU ESP8266 controller and an LCD display. A single indoor plant (money plant, Epipremnum aureum) is studied under light and dark conditions and at three light intensities.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

The study of gas exchange in plants has a long history. Blackman (1905) proposed the concept of limiting factors, showing that the rate of photosynthesis is controlled by whichever factor, such as light, CO_2 or temperature, is in shortest supply [1]. Farquhar, von Caemmerer and Berry (1980) later developed a biochemical model of CO_2 assimilation in C3 leaves that is still the basis of most modern photosynthesis research [2]. Standard textbooks on plant physiology explain stomatal control and the relation between assimilation and intercellular CO_2 [3].

Free-air CO_2 enrichment (FACE) experiments reviewed by Long et al. (2004) and Ainsworth and Long (2005) showed that increased CO_2 raises photosynthesis in many C3 plants, although the benefit is reduced by nutrient limitation and acclimation [4], [5]. These studies relied on expensive infrared gas analysers and field infrastructure. The Intergovernmental Panel on Climate Change reports document the rise in atmospheric CO_2 and stress the importance of terrestrial vegetation as a carbon sink [6].

On the instrumentation side, the Internet of Things (IoT) has enabled distributed, low-cost sensing [7]. The ESP8266 Wi-Fi microcontroller [8] and NDIR CO_2 modules such as the Winsen MH-Z19B [9] are widely used in air-quality monitors, smart agriculture and greenhouse control. Several student and hobby projects use these modules to log indoor CO_2, but most of them monitor only the air of a room and do not connect the measurements to a living plant in a controlled chamber.

2.2 Research Gap

From the literature it is clear that (i) precise gas-exchange measurements are available mainly through costly commercial analysers, (ii) low-cost IoT air-quality monitors usually record CO_2 alone, and (iii) very few educational prototypes measure CO_2 uptake

little documentation of how such a system can be built, calibrated and interpreted by undergraduate students. This project addresses these gaps by providing a complete, low-cost and reproducible experimental set-up.

2.3 Objective

The main objective of the project is to design and fabricate a low-cost IoT-based experimental system that measures, displays and analyses the carbon dioxide uptake and oxygen release of a plant under different light conditions, so that the contribution of plants to atmospheric balance can be demonstrated quantitatively. The detailed objectives are listed in Chapter 4.2.4 Flowchart

The flow of the work carried out in this project, from literature survey to conclusion, is shown in Figure 2.1.

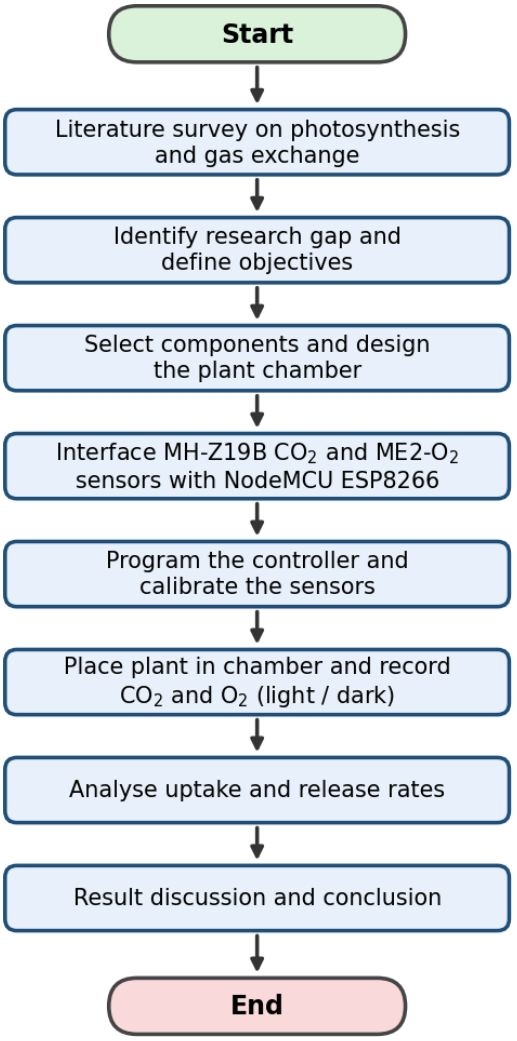


Figure 2.1: Flowchart of the project work

CHAPTER 3
PROBLEM IDENTIFICATION

The following problems motivated the present work. Each of them was identified through a literature survey and through observations made during the EVS course.

3.1 Rising Atmospheric Carbon Dioxide

Global CO_2 concentration has crossed 420 ppm and continues to rise every year. Plants are the principal natural sink, but their actual contribution is rarely quantified by students.

The concentration of carbon dioxide in the atmosphere has increased from about 280 ppm before the industrial revolution to more than 420 ppm at present, and the curve continues to rise every year. The main cause is the combustion of coal, oil and natural gas for electricity, transport and industry, supplemented by cement production and the clearing of forests. Because CO_2 absorbs infrared radiation emitted by the Earth's surface, a higher concentration traps more heat and contributes to global warming, melting of glaciers, sea-level rise and more frequent extreme weather events [6].

Plants, together with the oceans, form the principal natural sinks of atmospheric carbon dioxide. During photosynthesis, a leaf takes in CO_2 and converts it into sugars, starch, cellulose and other organic compounds that are stored in leaves, stems, roots and soil. Terrestrial vegetation therefore removes a large part of the carbon that human activities add to the air each year. Despite this, the contribution of an individual plant or a small group of plants is almost never measured by students, and it remains a figure quoted in textbooks rather than a result observed in the laboratory.

This lack of direct measurement is the first problem addressed by the present work. A student who sees a graph of rising global CO_2 naturally asks how much carbon dioxide a single houseplant can remove from the air around it, and under what conditions it does so. Answering that question requires a sealed chamber, a reliable CO_2 sensor and a record of concentration against time. The project is therefore designed to convert the

abstract idea of a carbon sink into a number in parts per million per minute that can be obtained, repeated and discussed in an EVS laboratory.

3.2 Decline of Green Cover

Urban expansion, mining and agriculture have reduced forest and green areas, lowering the capacity of the biosphere to absorb CO_2 and to regenerate oxygen.

Forests, grasslands, wetlands and urban trees are the living infrastructure through which the biosphere absorbs carbon dioxide and regenerates oxygen. Over the past century a very large area of natural vegetation has been converted to other uses. Agriculture is the largest driver of this change, followed by mining, road building, industrial development and the rapid expansion of cities. In many Indian towns, avenue trees are felled for road widening and open land is replaced by concrete, so that the green cover per person keeps falling even as the population grows.

The loss of green cover has two linked consequences. First, the quantity of CO_2 that can be removed by photosynthesis decreases, because there are fewer leaves to carry out the process. Second, when forests are cut or burned, the carbon stored in the wood and soil is released to the air as CO_2, so that a former sink turns into a source. Local effects such as higher surface temperature, poorer air quality, loss of habitat and reduced groundwater recharge accompany this global effect and make the problem visible even in a college campus or a residential neighbourhood.

3.3 Poor Awareness of the Role of Plants

Many people know that plants “give oxygen”, but they have no sense of the magnitude involved or of the fact that plants also release CO_2 at night. Evidence-based demonstrations are needed to correct such misconceptions.

Environmental awareness among the general public is often limited to slogans such as “plants give us oxygen” or “plant more trees”. While these statements are correct in outline, they hide several important points. Most people have no idea how much oxygen a plant actually releases, how many plants would be needed to influence the air in a

room, or how the process depends on light. As a result, expectations about indoor plants are often unrealistic, and the real benefits of large-scale tree cover are underestimated.

A common misconception is that plants release oxygen at all times. In reality, photosynthesis requires light, whereas respiration continues in every living cell throughout the day and night. In darkness a plant therefore consumes oxygen and releases carbon dioxide, just as animals do. The popular advice to avoid keeping plants in a bedroom at night is partly based on this fact, but it is rarely supported by numbers, and the quantities involved are in fact very small compared with the CO_2 released by the sleeping person.

Misconceptions of this kind are best corrected by evidence, not by repeating statements from textbooks. A demonstration in which learners watch the CO_2 reading fall when the lamp is switched on, and rise again when the chamber is darkened, leaves a much stronger impression than a lecture. The present project is intended to supply this type of evidence-based demonstration and thereby to improve the understanding of the role of plants among students and, through them, among the wider community.

3.4 Gas Exchange Is Invisible

CO_2 and O_2 are colourless and odourless gases, so photosynthesis cannot be observed directly. Without instruments, classroom explanations remain theoretical.

Both carbon dioxide and oxygen are colourless, odourless and tasteless gases, and the changes in their concentration during photosynthesis are far too small to be sensed by a human observer. A plant sitting in sunlight looks exactly the same as a plant sitting in shade, even though the amount of gas it exchanges with the air may differ several times. Photosynthesis is therefore an invisible process, and its effect on the atmosphere can be demonstrated only through measurement.

In many classrooms the process is explained with a chemical equation and a diagram of a leaf. Students may memorise the equation and the names of the organelles involved, yet they never see an observable effect of the reaction. This gap between theory and observation makes the topic abstract and weakens conceptual understanding.

Experiments that give an indirect visual cue, such as the starch test on a leaf or bubbles rising from an aquatic plant, show only that photosynthesis has occurred; they do not give a quantitative, continuous record of the gases.

Electronic gas sensors offer a remedy because they translate concentration into an electrical signal that can be read, displayed and plotted. When a CO_2 value on an LCD screen drops steadily while a lamp is on, the invisible process becomes visible in the form of numbers and graphs. The challenge identified here is to select sensors that are sensitive enough to detect the small changes occurring in a closed chamber and to present them in a form that learners can easily interpret.

3.5 High Cost of Standard Instruments

Commercial portable photosynthesis systems and infrared gas analysers are expensive and are normally unavailable in undergraduate laboratories.Scientific instruments that measure photosynthesis accurately, such as portable photosynthesis systems with infrared gas analysers, are designed for research laboratories and field stations. A single unit may cost several lakhs of rupees, and it often requires calibration gases, regular servicing and trained operators. Such equipment is beyond the budget of most undergraduate colleges, and even when one unit is available, it can be used by only one group of students at a time.

The high cost has a direct effect on teaching. Instead of performing the measurement themselves, students are usually shown a demonstration or a recorded result, and the opportunity to design, assemble and troubleshoot an experiment is lost. The financial barrier also limits repeated trials, which are essential in any experiment in biology because the behaviour of living material varies from one individual and one day to another.

The present work addresses this problem by building the measuring system from low-cost, widely available modules: a NodeMCU ESP8266 board, an MH-Z19B NDIR carbon dioxide sensor, an ME2-O_2 oxygen sensor, an LCD display and an acrylic chamber. The total cost of these parts is only a small fraction of the price of a

commercial analyser. Although the accuracy of the set-up is lower than that of research-grade instruments, it is adequate for demonstrating trends, for comparing conditions and for teaching the principles of gas-exchange measurement.

3.6 Errors in Manual and Chemical Methods

Traditional methods such as the use of indicator solutions, Winkler titration or counting oxygen bubbles from aquatic plants are slow, subjective and not suitable for continuous measurement.

Before electronic sensors became affordable, photosynthesis and respiration were studied through manual and chemical methods. Indicator solutions such as bicarbonate or phenol-red indicator change colour as CO_2 is removed from or added to water. Winkler titration determines dissolved oxygen by a series of chemical reactions followed by titration with sodium thiosulphate. Another classical activity is to count the bubbles of oxygen released per minute by an aquatic plant such as Hydrilla placed under a funnel in a beaker of water.

These methods are useful for introducing the concept, but they have clear limitations. Colour changes are judged by eye, so the endpoint differs from one observer to another. Titration needs careful handling of reagents, is time-consuming and gives only a single value for each sample. Bubble counting depends on the size of the bubbles and on the attention of the observer, and it is applicable only to submerged plants and not to terrestrial plants growing in air.

3.7 Lack of Real-Time Data

Without automatic data logging, short-term trends such as the response of a plant to a sudden change in light cannot be recorded or plotted. A plant responds to changes in its surroundings within seconds or minutes. When the light intensity is increased, the stomata open further, the rate of CO_2 uptake rises and the concentration of oxygen in the surrounding air begins to increase. When the light is switched off, the uptake falls rapidly and respiration becomes dominant.

Without automatic data logging, the experimenter is forced to read an instrument, write the value in a notebook and move to the next reading. This is not only tedious but also introduces mistakes in transcription and timing. The number of readings that can be collected is small, and the resulting graph does not have enough points to reveal fine details such as the initial lag, the steady-state slope or the point of saturation.

The absence of real-time data also reduces the interest of the learner, because the result is known only after the experiment is over. In the proposed system the microcontroller reads both sensors continuously, shows the values on the LCD and can transmit them through the Wi-Fi module of the ESP8266 to a computer for plotting. Real-time observation allows the experimenter to notice problems such as sensor drift or a leak in the chamber immediately and to correct them during the experiment instead of after it.

3.8 Variation with Species and Environmental Conditions

The rate of gas exchange differs between species and changes with light, temperature, humidity and enclosure volume. A flexible set-up is needed to study these variations.The rate at which a plant exchanges gases is not a fixed property. It differs between species, between C3 and C4 plants, between young and mature leaves, and even between two leaves of the same plant. It also depends strongly on the environment. According to the principle of limiting factors, the rate of photosynthesis is controlled by whichever factor, such as light intensity, CO_2 concentration or temperature, is in shortest supply [1]. Humidity influences the opening of the stomata, and water stress can close them completely and stop gas exchange.

The volume of the enclosure is another variable, because the same amount of CO_2 taken up by the plant produces a larger change in concentration in a small chamber than in a large one. The temperature inside a sealed transparent chamber also rises under a lamp, which affects both the plant and the sensor readings.

A suitable experimental system must therefore be flexible. It should allow the light intensity to be changed in steps, should permit different plants and pot sizes to be placed in the chamber, and should include a temperature sensor so that thermal effects can be separated from biological ones. Control experiments with an empty chamber and with a pot containing only soil are also necessary to account for leakage, soil respiration and sensor limitations. The design adopted in this project takes these requirements into consideration.

3.9 Need for Educational Demonstrators

Environmental education benefits from simple working models that learners can build, operate and modify themselves, combining sensors, electronics and biology. Environmental education is most effective when learners take part in activities rather than merely listen to explanations. Working models that students can build, operate, modify and repair themselves develop practical skills and a feeling of ownership of the knowledge. They also encourage interdisciplinary thinking, because a gas-exchange experiment combines plant biology, chemistry of gases, electronics, programming and data analysis.

At present, ready-made educational kits for studying plant gas exchange are rare, and the few that exist are either costly or provide only qualitative information. Many projects on microcontrollers and sensors, on the other hand, are not connected with a real scientific question and remain exercises in wiring. There is a need for demonstrators that join the two fields, so that electronics is used to answer a biological question and the biological question gives a purpose to the electronics.

3.10 Problem Statement

There is a need for a compact, affordable and reliable experimental system that can continuously measure CO_2 uptake and O_2 release of a plant, display the values in real time and allow the results to be analysed under different conditions.

The problems discussed in the previous sections can be summarised in one statement. The rising level of CO_2 and the loss of green cover make the role of plants in maintaining atmospheric balance more important than ever, but public understanding of this role is superficial, the process itself is invisible, and the instruments that can quantify it are costly and rarely available to students. The older manual methods are slow and subjective, and they cannot provide real-time information.

Table 3.1 relates each of the problems identified in this chapter to the approach adopted in the project in order to address it. It shows that a single, compact experimental system, consisting of a sealed plant chamber, two gas sensors, a microcontroller and a display, can respond to all of the problems at the same time. The chapter that follows converts this response into a set of specific objectives.

Table 3.1: Problems identified and the approach adopted in the project

S.No.	Problem identified	Approach adopted in the project
1	Rising atmospheric CO_2	Measure the CO_2 uptake of a living plant in ppm per minute
2	Decline of green cover	Quantify the ecosystem service provided by vegetation
3	Poor awareness of plants' role	Provide evidence-based, light and dark demonstrations
4	Invisible gas exchange	Convert gas concentration into displayed numbers and graphs
5	High cost of instruments	Use low-cost NDIR and electrochemical sensor modules
6	Errors in manual methods	Use automatic, objective sensor readings
7	Lack of real-time data	Continuous logging and display with NodeMCU ESP8266
8	Species and condition variation	Flexible chamber, variable light and temperature monitoring
9	Need for educational demonstrators	Reusable, open-source and easy-to-build prototype

CHAPTER 4
OBJECTIVES

4.1 Study of Plant Gas Exchange

To study the processes of photosynthesis and respiration and to understand how they govern the uptake of CO_2 and the release of O_2 in green plants.The first objective is to build a sound understanding of how green plants exchange gases with their surroundings. This includes the structure of the leaf and the role of the stomata, the light reactions in which water is split and oxygen is released, and the Calvin cycle in which CO_2 is fixed into carbohydrates. It also includes the process of respiration, in which sugars are oxidised to release energy, carbon dioxide and water, and which operates in all living cells throughout the day and night.

The study also aims to explain why the gas exchange measured around a plant is a net quantity. In light, photosynthesis exceeds respiration and the plant takes up CO_2 and releases O_2 overall. At the compensation point the two processes balance, and in darkness respiration alone remains, so the plant releases CO_2 and consumes O_2. Understanding this relationship is necessary for interpreting the sensor readings correctly in the later stages of the project.

Another aim within this objective is to identify the factors that affect the rate of gas exchange, namely light intensity, temperature, humidity, CO_2 concentration, water supply and the health of the plant. The information is gathered from textbooks, research papers and standard reference sources’

4.2 Design of the Plant Chamber

To design a transparent, sealed chamber of 300 mm × 300 mm × 350 mm in which a potted plant can be kept and the air composition can be measured without outside interference.

The second objective is to design and construct a chamber in which a potted plant can be kept and the composition of the air can be monitored without interference from the surroundings. The chamber is made of transparent acrylic sheet with internal dimensions of 300 mm × 300 mm × 350 mm, giving a volume of about 31.5 litres. Transparency is required so that light from the external LED source reaches the leaves, and acrylic is chosen because it is light, strong, easy to cut and does not absorb the gases of interest.

The chamber must be gas-tight so that changes in concentration can be attributed to the plant and not to exchange with the room air. For this reason all joints are sealed with a suitable sealant, and the openings for the sensor wires are closed carefully. A small fan can be used to mix the air so that the sensors are exposed to a uniform sample, and the electronics are kept outside the humid enclosure as far as possible to protect them from moisture.

The size of the chamber is a compromise. It should be large enough to hold a small plant such as a money plant without crowding the leaves, but small enough that the plant produces a measurable change in concentration within a reasonable time. A leak test using the CO_2 sensor with an empty chamber is planned to confirm that the design meets the requirement before any plant is introduced.

4.3 Development of the Sensing System

To interface an NDIR carbon dioxide sensor and an electrochemical oxygen sensor with a NodeMCU ESP8266 development board and to display the readings on an LCD.The third objective is to develop the sensing and display system that measures and shows the gas concentrations. An MH-Z19B non-dispersive infrared sensor is used for carbon dioxide because of its selectivity and long-term stability, and an ME2-O_2 electrochemical sensor is used for oxygen because it gives an output proportional to the oxygen partial pressure. Both sensors are connected to a NodeMCU ESP8266 development board, which serves as the central controller of the system.

The CO_2 sensor communicates with the controller through a serial (UART) interface, while the output of the oxygen sensor is amplified and read through the analogue input of the NodeMCU. A 16 × 2 LCD with an I2C module is added so that the CO_2 concentration in ppm and the O_2 concentration in per cent can be read directly on the apparatus. The program is written in the Arduino IDE and includes functions for reading the sensors, converting the raw values into engineering units and updating the display at regular intervals.

Calibration is an essential part of this objective. The oxygen sensor is calibrated against fresh outdoor air at 20.9 % O_2, and the CO_2 sensor is checked against ambient air whose concentration is known approximately. The sensor readings are also compared before and after warm-up so that the settling time is known. The result is a working data-acquisition unit that can be used for all the later experiments.

4.4 Experimental Measurement

To measure the variation of CO_2 and O_2 concentration with time under light and dark conditions and at different light intensities.The fourth objective is to carry out the experimental measurements. The plant is placed in the sealed chamber, the sensors are allowed to stabilise, and the concentrations of CO_2 and O_2 are recorded at fixed intervals. The measurement is repeated under several conditions: with the lamp switched on, with the lamp switched off, and at three different light intensities obtained by changing the power or the distance of the LED source.

Control experiments are included to make the results reliable. A run with an empty chamber shows the leakage rate and the drift of the sensors, and a run with the pot and soil but without the plant shows the contribution of soil microorganisms to the gas exchange. The temperature inside the chamber is noted during every run, since the heat from the lamp can alter both the plant’s response and the sensor output. Each experiment is repeated at least three times so that the variation between trials can be estimated.

4.5 Analysis of Results

To calculate the net CO_2 uptake rate in ppm per minute and in µmol m^{-2} s^{-1}, to compare it with O_2 release and to discuss the findings.The fifth objective is to analyse the recorded data and to draw meaningful conclusions. The readings are plotted as graphs of CO_2 concentration and O_2 concentration against time, and the steady portion of each curve is fitted with a straight line. The slope of this line gives the net rate of change of CO_2 in ppm per minute, and the corresponding slope for oxygen gives the rate of change of O_2 in per cent per minute.

To compare the results with values reported in the literature, the rate in ppm per minute is converted into the standard unit of µmol m^{-2} s^{-1}. This conversion uses the volume of the chamber, the temperature and pressure of the air, and the leaf area of the plant. The oxygen release is then compared with the CO_2 uptake. If the system works properly, the number of moles of O_2 released should be close to the number of moles of CO_2 absorbed, which is the 1 : 1 ratio predicted by the photosynthetic equation.

Differences between the expected and observed values are discussed in terms of possible sources of error, such as leakage, sensor drift, the response time of the sensors and the effect of respiration. The analysis also compares the three light intensities to show whether the uptake increases with light and whether it approaches saturation. The findings are presented in the results and discussion chapter of this report.

4.6 Awareness and Sustainability

To create a low-cost model that promotes environmental awareness, supports sustainable development goals and can be reused for further EVS experiments.The sixth objective looks beyond the experiment itself and aims to create a model that promotes environmental awareness and supports sustainability. By presenting the performance of a living plant in numerical form, the system helps learners to appreciate the contribution of vegetation to the control of atmospheric CO_2 and to the supply .

The project is also linked with several of the United Nations Sustainable Development Goals, in particular SDG 4 (Quality Education), SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action) and SDG 15 (Life on Land). It encourages the use of low-cost technology and open-source tools, which makes advanced environmental measurement accessible to institutions with limited resources.

Finally, the apparatus is designed to be reusable. The same chamber, controller and sensors can be used for experiments on different plant species, on the effect of pollutants or on indoor air quality, and a temperature sensor, a light sensor or a cloud dashboard can be added later. Table 4.1 shows how each objective is connected with the method followed and the outcome expected, and it summarises the plan of the work reported in the following chapters.

Table 4.1: Relation between objectives, methods and expected outcomes

Obj.	Method followed	Expected outcome
4.1	Literature survey and study of photosynthesis and respiration	Theoretical basis for the experiment
4.2	Fabrication of 300 × 300 × 350 mm sealed acrylic chamber	Gas-tight chamber with controlled light
4.3	Interfacing MH-Z19B and ME2-O_2 sensors with NodeMCU and LCD	Working, calibrated data-acquisition unit
4.4	Recording CO_2 and O_2 in light and dark at three intensities	Time-dependent data set with controls
4.5	Slope calculation, unit conversion and O_2 : CO_2 comparison	Net uptake rate and discussion of errors
4.6	Documentation and linking with SDGs	Reusable, awareness-building model

CHAPTER 5

COMPONENTS USED

5.1 Introduction

The hardware of the system is built around a small number of readily available components. The microcontroller reads the two gas sensors, processes the data and shows them on an LCD. Power and programming are provided through a USB cable. This chapter describes each component and its role in the project.

5.2 Components Used

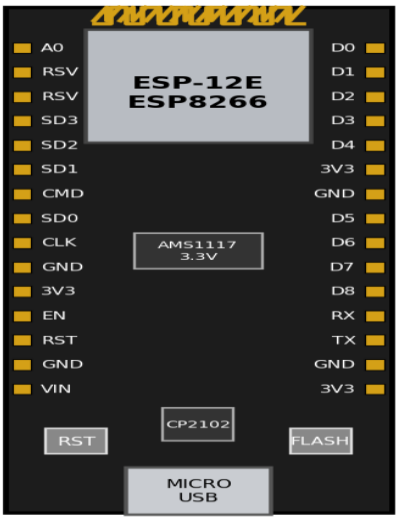
Table 5.1: List of components used

S.No.	Component	Specification	Qty
1	NodeMCU ESP8266	32-bit CPU, Wi-Fi, 3.3 V logic	1
2	16 × 2 LCD with I2C module	5 V, I2C address 0x27	1
3	ME2-O_2 oxygen sensor	Electrochemical, 0–25 % O_2	1
4	MH-Z19B CO_2 sensor	NDIR, 0–5000 ppm, UART	1
5	USB Type-B cable	Power and programming, 5 V	1
6	Acrylic chamber and LED light	300 × 300 × 350 mm; 12 W LED	1 set
7	Jumper wires, breadboard	Male/female	1 set

5.3 NodeMCU ESP8266 Development Board

The NodeMCU is an open-source development board built on the ESP-12E module with the ESP8266 system-on-chip. It operates at 3.3 V, runs at 80/160 MHz, has 4 MB of flash memory, 11 usable GPIO pins, one 10-bit analogue input (A0) and an integrated Wi-Fi transceiver. A CP2102 USB-to-serial chip allows programming through the Arduino IDE. In this project it acts as the central controller.

The processed data is displayed on the LCD display, allowing the user to observe changes in CO_2 uptake and oxygen release in real time. The NodeMCU ESP8266 also supports Wi-Fi connectivity, which can be used to transmit experimental data to an online platform for storage and analysis.



NodeMCU ESP8266 Development Board

Figure 5.1: NodeMCU ESP8266 development board

5.4 LCD Display

A 16 × 2 character liquid crystal display shows the live CO_2 concentration (ppm) and O_2 concentration (%). An I2C backpack based on the PCF8574 chip reduces the number of required connections to four (GND, VCC, SDA and SCL)

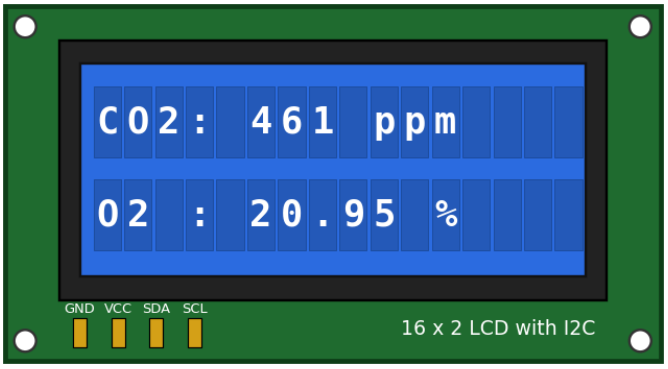


Figure 5.2: 16 × 2 LCD display with I2C module

5.5 Oxygen Sensor

The ME2-O_2 is an electrochemical (galvanic fuel-cell) sensor. Oxygen diffusing through a membrane is reduced at the cathode, producing a current proportional to the oxygen partial pressure. With a signal-conditioning circuit, the output voltage of about 0.1 V in normal air is amplified and fed to the analogue pin A0 of the NodeMCU. Typical measuring range is 0–25 % O_2.

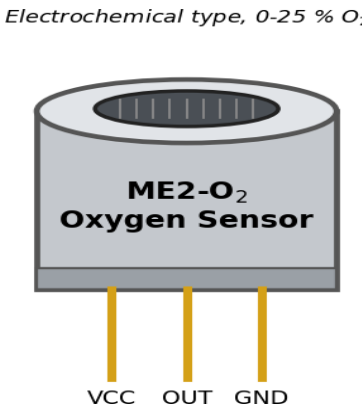


Figure 5.3: ME2-O_2 electrochemical oxygen sensor

5.6 CO_2 Sensor

The MH-Z19B is a compact NDIR sensor with a gold-plated gas chamber. Infrared light of about 4.26 µm is absorbed by CO_2 molecules, and the reduction in intensity gives the concentration. It measures 0–5000 ppm with an accuracy of ±(50 ppm + 5 % of reading), operates on 4.5–5.5 V and provides a UART (9600 baud) and a PWM output. Its selectivity and long-term stability make it suitable for plant studies.

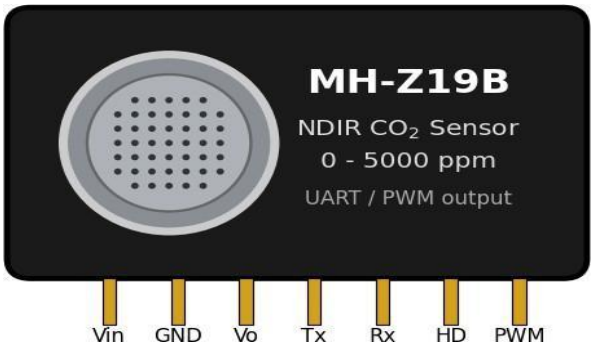


Figure 5.4: MH-Z19B NDIR carbon dioxide sensor

5.7 USB Type-B Cable

A USB Type-B cable connects the controller to a computer or to a 5 V adapter. It supplies power to the NodeMCU, the LCD and the two sensors, and it is used for uploading the program and for viewing the data on the serial monitor.



Figure 5.5: USB Type-B cable

CHAPTER 6

SYSTEM DESIGN AND METHODOLOGY

6.1 Design Considerations

The system was designed with the following considerations: (i) the chamber must be transparent and gas-tight, (ii) the sensors must be exposed to well-mixed chamber air,(iii) the electronics must remain outside the humid chamber as far as possible.

6.2 Block Diagram

The block diagram in Figure 6.1 shows that the CO_2 sensor communicates with the NodeMCU through UART, the O_2 sensor through the analogue input, and the LCD through the I2C bus.

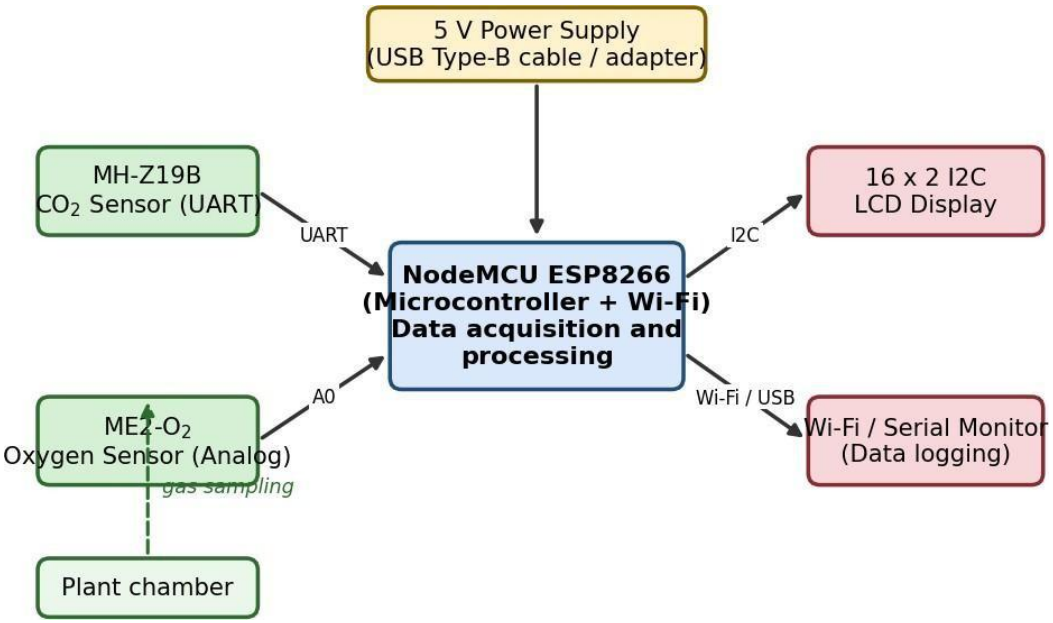


Figure 6.1: Block diagram of the system

6.3 Circuit Connections

Module	Module pin	NodeMCU pin
MH-Z19B	Vin / GND	VIN (5 V) / GND
MH-Z19B	Tx	D6 (GPIO12)
MH-Z19B	Rx	D5 (GPIO14)

Module	Module pin	NodeMCU pin
ME2-O_2 module	VCC / GND	3V3 / GND
ME2-O_2 module	OUT	A0
LCD I2C	VCC / GND	VIN (5 V) / GND
LCD I2C	SDA	D2 (GPIO4)
LCD I2C	SCL	D1 (GPIO5)

6.4 Experimental Set-up

The potted plant is placed at the centre of the acrylic chamber. The two sensors are fixed on the inner wall at mid-height, and a small fan is optionally used to mix the air. The cable entries and the lid are sealed with silicone. An LED grow light fixed under the roof provides controlled illumination, and a lux meter is used to set the light intensity. Experiments are run in light and dark conditions, each for 60 minutes.

6.5 Software and Programming

The program was written in the Arduino IDE using the ESP8266 board package. The algorithm is: initialise the serial port, LCD and sensors; wait for the CO_2 sensor warm-up; read the CO_2 concentration from the UART frame; read the analogue voltage of the O_2 sensor and convert it to percentage; display both values on the LCD; send them to the serial monitor along with a time stamp every 10 seconds; and repeat. Average of ten readings is taken to reduce noise.

6.6 Calibration and Calculation

The O_2 sensor is calibrated in fresh outdoor air, which is taken as 20.9 % O_2. The oxygen percentage is calculated as O_2 (%) = (V / V_air) × 20.9, where V is the measured output voltage and V_air is the voltage in fresh air. The MH-Z19B is zero-calibrated at 400 ppm in outdoor air. The net CO_2 uptake rate is calculated from the change in concentration:

CHAPTER 7
DESIGN MODEL

7.1 Design Overview

The design model consists of a cuboid acrylic chamber with a base plate, an LED light fixed below the roof, a side-mounted electronics box and a sealed lid. Acrylic was selected because it is transparent, lightweight, easy to cut and cement, and it does not react with the gases being measured. The effective air volume of the chamber after subtracting the pot is about 30 litres.

Table 7.1: Main design parameters

Parameter	Value
Chamber size	300 × 300 × 350 mm
Wall thickness	4 mm acrylic
Effective volume	≈ 30 L
Light source	12 W LED grow light
Supply	5 V, 1 A through USB

The working model connects the sensors to the microcontroller, which collects and processes the measured data. The experimental readings can be displayed on an LCD or computer and recorded for further analysis. Initially, the gas concentrations are measured before exposing the plant to the selected light conditions. After illumination, the changes in carbon dioxide and oxygen concentrations are observed at regular intervals. The recorded data is compared to evaluate the relationship between carbon dioxide uptake and oxygen release.

The design allows experiments under different light intensities and temperatures, helping to understand how environmental conditions affect photosynthesis. This model provides a simple, sensor-based method for studying plant gas exchange and supporting environmental monitoring and agricultural research.

7.2 2D VIEW

Figure 7.1 gives the front and top views of the chamber with the main dimensions. The sensors are mounted on the right-hand wall at different heights, and the electronics box with the LCD is kept outside so that the display can be read without opening the

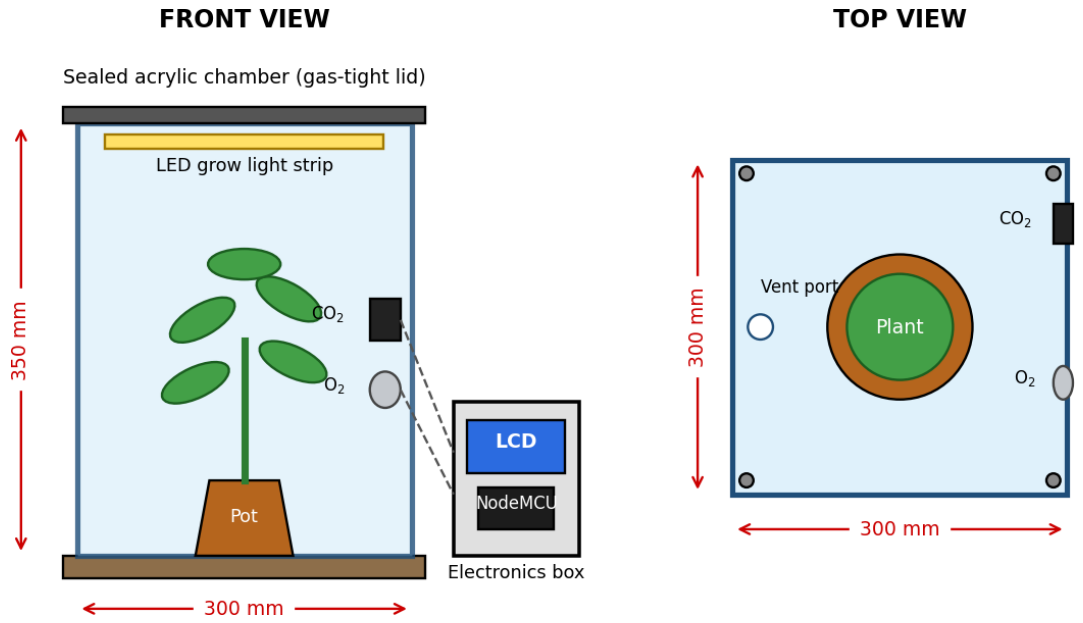


Figure 7.1: 2D design of the plant chamber (front and top views)

7.3 3D VIEW

The 3D model in Figure 7.2 shows the assembled system. The transparent walls allow light to reach the plant, the sensors are located close to the back wall.

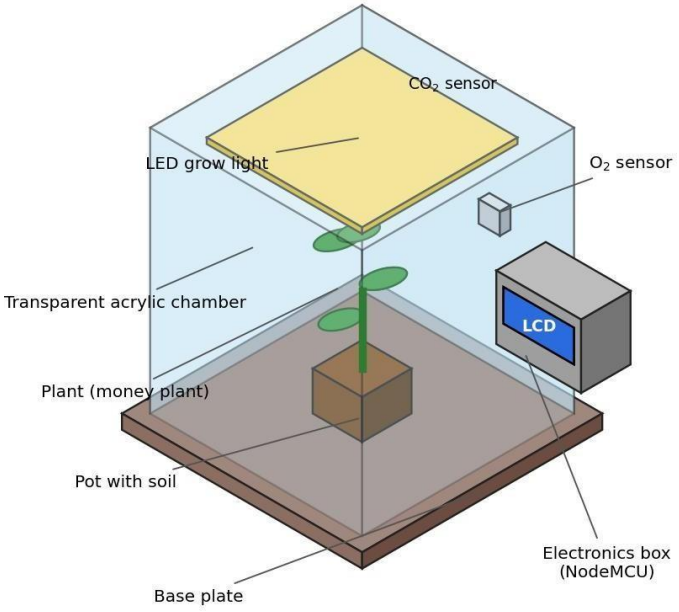


Figure 7.2: 3D model of the experimental set-up

CHAPTER 8
FABRICATION AND IMPLEMENTATION

8.1 Selection of Materials

Transparent 4 mm acrylic sheets were chosen for the chamber walls, and a 10 mm plywood sheet was used for the base. Silicone sealant gave an airtight joint, and a PVC box was used to house the electronics. All components were procured from local electronics suppliers.

8.2 Wi-Fi-Based Real-Time Monitoring System

The proposed project uses the NodeMCU ESP8266 Wi-Fi module to monitor carbon dioxide (CO_2) uptake and oxygen (O_2) release in plants during photosynthesis. The CO_2 sensor and oxygen sensor collect gas concentration readings, which are processed by the NodeMCU microcontroller. Using its built-in Wi-Fi capability, the NodeMCU can host a web server or transmit sensor readings to a suitable IoT platform. The readings can then be accessed through a website using a smartphone, laptop, or computer connected to the same network or an appropriate internet-connected service.

8.3 Web-Based Display of Sensor Readings

A web-based monitoring page can display the measured CO_2 and O_2 concentrations in a clear, organized format. The page can be programmed to refresh the readings automatically at regular intervals, allowing users to observe changes in gas concentrations under different lighting conditions. The collected data can also be recorded for comparison and further analysis.

8.4 Programming, Testing and Implementation

After uploading the program, the sensors were allowed to stabilise for three minutes. The O_2 sensor was calibrated in fresh air and the CO_2 sensor was checked against ambient air (about 420 ppm). A leak test was done by raising the CO_2 level to 1500 ppm and checking that it fell by less than 10 ppm in one hour with no plant inside. A money plant with a leaf area of about 250 cm^2 was then placed in the chamber and the experiments were performed.

CHAPTER 9

COST ESTIMATION

9.1 Electronic Components Cost

Table 9.1: Cost of electronic components

S.No.	Item	Cost (₹)
1	NodeMCU ESP8266	350
2	16 × 2 LCD with I2C	250
3	ME2-O_2 oxygen sensor	1800
4	MH-Z19B CO_2 sensor	2200
5	USB Type-B cable	120
6	5 V adapter	250
7	Jumper wires and breadboard	200
	Sub-total	5170

9.2 Fabrication and Miscellaneous Cost

Table 9.2: Cost of fabrication and miscellaneous items

S.No.	Item	Cost (₹)
1	Acrylic sheets and cutting	900
2	LED grow light	400
3	Silicone, gasket, PVC box	300
4	Plant and pot	150
	Sub-total	1750

9.3 Total Cost

Table 9.3: Total project cost

Description	Cost (₹)
Electronic components	5170
Fabrication and miscellaneous	1750
Total estimated cost	**6920**

The total cost of the prototype is approximately ₹6920, which is a small fraction of the price of a commercial photosynthesis measurement system. The cost can be reduced further by using a single sensor board or by reusing materials available in the laboratory.

CHAPTER 10
WORKING PRINCIPLE

10.1 Principle of Photosynthesis and Respiration

In light, chloroplasts use light energy to fix CO_2, so the CO_2 concentration inside a closed chamber decreases and the O_2 concentration increases. In darkness only respiration occurs, so the CO_2 concentration rises and O_2 falls slightly. The system measures these changes and converts them into rates.

10.2 Carbon Dioxide Sensing

The MH-Z19B works on the NDIR principle. An infrared lamp sends radiation through the gas chamber and a detector with a 4.26 μm filter measures the intensity. According to the Beer–Lambert law, the absorbed energy is proportional to the CO_2 concentration. The internal processor converts it to ppm and sends it as a serial data frame to the NodeMCU.

10.3 Oxygen Sensing

The ME2-O_2 sensor is a galvanic cell in which oxygen is reduced at the cathode while the anode is oxidised. The resulting current produces a voltage across a load resistor that is proportional to the oxygen partial pressure. After amplification, the NodeMCU reads this voltage with its ADC and converts it to a percentage using the calibration in Section 6.6.

10.4 Operation of the System

When the system is switched on, the NodeMCU initialises the sensors and the LCD. It then reads CO_2 and O_2 every few seconds, averages the values and displays them. The data are also transferred to the computer, where the change in concentration with time is plotted. The net uptake rate is calculated for each light condition, and the experiment is repeated for dark and for different light intensities. The resulting current produces a voltage across a load resistor that is proportional to the oxygen partial pressure.

CHAPTER 11
RESULT AND DISCUSSION

11.1 Experimental Observations

The experiments were performed with a money plant in the chamber at 28 °C. The chamber air was first enriched to about 1500 ppm CO_2 and the readings were noted at 10-minute intervals for 60 minutes in light (10,000 lux) and in darkness. The observations are given in Table 11.1. **Note:** the values are representative readings and must be replaced by the actual readings recorded by the student.

Table 11.1: CO_2 and O_2 concentration with time

Time (min)	CO_2 light (ppm)	O_2 light (%)	CO_2 dark (ppm)	O_2 dark (%)
0	1500	20.90	1500	20.90
10	1413	20.91	1520	20.90
20	1330	20.92	1541	20.90
30	1250	20.92	1560	20.89
40	1175	20.93	1579	20.89
50	1105	20.94	1597	20.89
60	1040	20.95	1616	20.89

11.2 Graphical Analysis

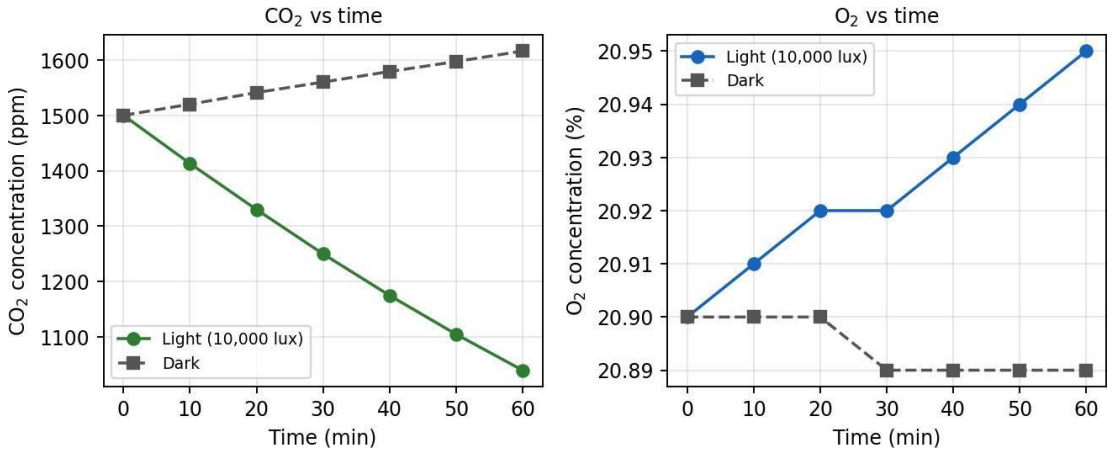


Figure 11.1: Variation of CO_2 and O_2 concentration with time

In light the CO_2 concentration fell from 1500 ppm to 1040 ppm in one hour, a net uptake of 460 ppm (7.67 ppm per minute), while the oxygen concentration rose by about 0.05%, which is close to the expected 1:1 ratio (460 ppm ≈ 0.046 %).

Condition	ΔCO_2 in 60 min (ppm)	Rate (ppm/min)	Rate (µmol m^{-2} s^{-1})
Dark	+116	−1.93	−1.58 (release)
Low light (3,000 lux)	−190	3.17	2.59
Medium light (6,000 lux)	−330	5.50	4.50
High light (10,000 lux)	−460	7.67	6.27

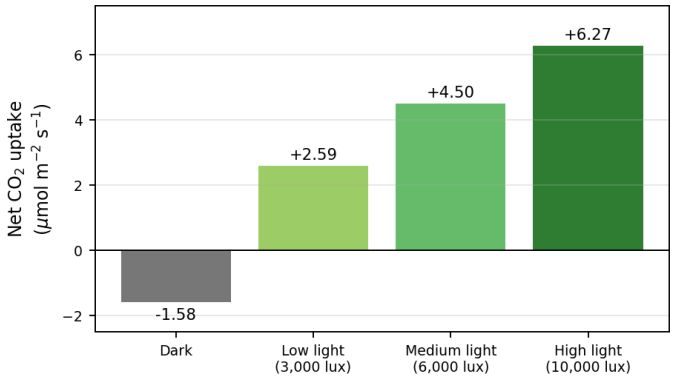


Figure 11.2: Net CO_2 uptake of the plant at different light intensities

11.3 Discussion

The results confirm that the plant acts as a net CO_2 sink in light and as a net source in darkness. The uptake increased with light intensity, showing the typical light-response behaviour of C3 plants, and the rate of 6.27 µmol m^{-2} s^{-1} at high light is within the range reported for indoor foliage plants. The calculation used V = 0.030 m^3, T = 298 K, P = 101325 Pa and A = 0.025 m^2.

The change in oxygen is small because the background oxygen level is high (20.9 %). Therefore, the O_2 readings are more sensitive to sensor resolution, temperature and humidity than the CO_2 readings, and the O_2 data should be treated as supporting evidence of the trend. Accuracy can be improved by temperature compensation, repeated trials and a sensor with higher resolution.

CHAPTER 12
CONCLUSION

The project “Experimental Analysis of Carbon Dioxide Uptake and Oxygen Release in Plants” provides an understanding of the photosynthesis process and the role of plants in maintaining atmospheric balance. Plants absorb carbon dioxide from the surrounding air and release oxygen during photosynthesis, supporting the survival of humans and other living organisms. This study helps explain the importance of plant activity in maintaining the natural balance of gases in the environment.

The proposed experimental system uses carbon dioxide and oxygen sensors, a temperature sensor, a light source, and an Arduino or ESP32 microcontroller to monitor gas concentration changes. The collected sensor readings can be recorded and analyzed to study the relationship between environmental conditions and photosynthetic activity. By changing light intensity and monitoring temperature, the experiment can help identify factors that influence carbon dioxide uptake and oxygen release. Accurate sensor calibration and controlled experimental conditions are essential for obtaining reliable results.

A low-cost IoT-based experimental system for measuring CO_2 uptake and O_2 release of plants was designed, fabricated and tested. The system uses a NodeMCU ESP8266, an MH-Z19B CO_2 sensor, an ME2-O_2 oxygen sensor and an LCD display inside a sealed acrylic chamber. The experiments showed that the plant absorbed CO_2 and released O_2 in light, and released CO_2 in darkness, and that the net uptake rate increased with light intensity.

In conclusion, this project demonstrates the usefulness of sensor-based technology for studying plant gas exchange and environmental conditions. The system can be further improved by adding wireless data transmission, real-time graphs, automated data logging, and cloud-based monitoring. It can be applied in educational laboratories, greenhouse management, agricultural research, and environmental studies.

CHAPTER 13
FUTURE SCOPE

1 The system can be developed for continuous monitoring of carbon dioxide uptake and oxygen release in plants under different environmental conditions.

2. Wireless communication can be integrated to transmit sensor readings to mobile phones and computers for remote monitoring.

3. The experimental setup can be upgraded with IoT technology to store and monitor plant gas exchange data in real time.

4. Advanced sensors can be used to improve the accuracy of carbon dioxide and oxygen measurements during photosynthesis.

5. Automatic control of light intensity, temperature, and humidity can be introduced to study suitable conditions for plant growth.

6. The system can be expanded to analyze different plant species and compare their photosynthesis performance under identical conditions.

7. Artificial intelligence can be implemented to analyze collected data and predict photosynthetic activity under varying environmental conditions.

8. The developed model can be applied in greenhouses, agricultural research laboratories, and environmental monitoring systems to support sustainable agriculture and carbon management.

In conclusion, this project demonstrates the usefulness of sensor-based technology for studying plant gas exchange and environmental conditions.

CHAPTER 14
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